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FACULTY OF APPLIED SCIENCES AND ENGINEERING
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

PROJECT TITLE:

**BORESHA-MAMA: MOBILE AND WEB-BASED MATERNAL HEALTHCARE SYSTEM
FOR TRANS-NZOIA COUNTY**

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SUBMISSION DATE: 27th FEBRUARY 2026

DECLARATION

I declare that this research project as presented in this paper is my original work and has not been presented anywhere else for any degree, diploma or award.

Signature: _____ Date: 27th FEBRUARY 2026

APPROVAL

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DEDICATION

This research project is dedicated to all expectant mothers and families in Trans-Nzoia County who face challenges in accessing timely and quality maternal healthcare. It is also dedicated to my loving family for their unwavering support, encouragement, and sacrifices throughout my academic journey.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisors, Dr. Sam Njuki and Mr. Luke Okelo, for their invaluable guidance, constructive feedback, and continuous support throughout this research project. Their expertise and encouragement are instrumental in shaping this project.

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Finally, I thank my family and friends for their moral and emotional support. This achievement would not have been possible without them. May God bless you all.

LIST OF ABBREVIATIONS

CHV - Community Health Volunteer

ICT - Information and Communication Technology

KNBS - Kenya National Bureau of Statistics

MOH - Ministry of Health

NCD - Non-Communicable Disease

NHIF - National Health Insurance Fund

SHA - Social Health Authority

WHO - World Health Organization

KSH - Kenyan Shillings

BP - Blood Pressure

UML - Unified Modeling Language

ANC - Antenatal Care

VS Code - Visual Studio Code

ER - Entity-Relationship

AWS - Amazon Web Services

PWA – Progressive Web Application

ROI – Return On Investment

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ABSTRACT

This research project presents Boresha-Mama, a mobile and web-based maternal healthcare system designed to improve maternal care delivery in Trans-Nzoia County, Kenya. The system connects mothers, Community Health Volunteers (CHVs), health facilities, and county officials to address challenges such as long-distance travel to clinics, loss of paper records, and missed antenatal check-ups. By enabling real-time data sharing, offline functionality, pregnancy tracking, home visit recording, and dashboards for planning, Boresha-Mama reduces preventable maternal and neonatal deaths through early risk detection and regular CHV follow-ups.

Maternal healthcare in rural Kenya faces significant barriers, including poor infrastructure and fragmented records. Boresha-Mama leverages accessible smartphone technology with offline-first design to bridge these gaps. Developed using Agile methodology, the project spans six months at an estimated cost of KSH 160,000. Following a pilot in Kiminini constituency, the system can scale to other counties, aligning with Kenya's Universal Health Coverage goals and Sustainable Development Goal 3.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

A maternal healthcare system is formally defined as the organised provision of health services to women during pregnancy, childbirth, and the postnatal period to ensure positive experiences and reduce preventable morbidity and mortality (World Health Organization, 2024).

The concept of organised maternal health services dates back to the early 20th century. In the United States, the Children's Bureau was established in 1912 by the federal government to promote maternal and child health, followed by the Sheppard-Towner Act of 1921, which supported state-level maternal and infant care programmes (Warren, 2023). The first major national maternal health programme in Kenya was the free maternity services policy introduced in all public health facilities in June 2013 by the Government of Kenya to eliminate user fees for maternal services (Orangi et al., 2021). This was expanded in 2017 into the Linda Mama programme under the National Hospital Insurance Fund (NHIF).

Since its introduction in 2013, maternal healthcare programmes such as Linda Mama have reached millions of women in Kenya. The 2022 Kenya Demographic and Health Survey (KDHS) indicates that nearly all women (98%) received antenatal care from a skilled provider, with substantial increases in facility-based deliveries (KNBS and ICF, 2023). Linda Mama remains the most popular and widely recognised maternal healthcare system in Kenya due to its nationwide coverage and direct impact on reducing financial barriers.

Maternal healthcare systems are commonly used in both urban and rural health facilities across Kenya, with higher utilisation in public and private sectors where services are subsidised or insured. However, rural areas like Trans-Nzoia County continue to experience challenges. According to the Kenya National Bureau of Statistics (KNBS) and Ministry of Health (MoH) reports, women in rural Kenya face long distances to facilities (often 50-100 km), poor roads, and high transport costs, leading to missed check-ups and elevated maternal mortality (KNBS and ICF, 2023; Ministry of Health, 2024). The World Health Organization notes that Kenya's maternal mortality ratio stood at approximately 355 deaths per 100,000 live births in recent estimates, with many preventable deaths linked to delayed care in rural settings (World Health Organization, 2024).

1.1.1 Overview of Existing System

The Linda Mama programme (2017–2024) provided free antenatal care, skilled delivery, postnatal care, and infant services up to nine months, significantly increasing utilisation of maternal services (Orangi et al., 2021). It helped reduce financial barriers for low-income mothers. However, as of October 2024, Linda Mama is no longer a standalone programme; it has been integrated into the broader Social Health Authority (SHA) under the Social Health Insurance framework, now part of expanded household-based coverage known as Linda Jamii (ThinkWell, 2025). Records remain largely paper-based in many facilities, with limited digital integration for CHVs and real-time county planning.

1.1.2 Overview of Proposed System

This research project proposes Boresha-Mama, a mobile and web-based system that uses technology to link mothers, CHVs, health facilities, and county officials, making them more connected in providing maternal healthcare and lowering child deaths and miscarriages via regular checks.

1.2 Problem Statement

Expectant women in Trans-Nzoia deal with big troubles. They go far to clinics, which wears them out. Paper records are unsafe for they disappear or get forgotten during visits to the health facilities. This slows finding issues like high blood pressure or sicknesses. Travel and visit costs are too much, so some skip check-ups. In rainy times due to poor roads, paths are blocked/inaccessible affecting access to the health facilities. This risks both the mother's and the baby's lives.

Current systems are not linked well. A mother sees one doctor, then another, but the patient information is not shared among the doctors affecting diagnosis. CHVs visit houses but record the findings on books, which take long to reach clinics. County health officials cannot see fresh data for better planning. This raises deaths of mothers and babies, plus miscarriages, as World Health Organization (WHO) notes for Kenya.

The core issue is the lack of a cohesive digital infrastructure to support maternal healthcare delivery in Trans-Nzoia County. This deficiency manifests in several interconnected problems: disconnected care pathway where information does not flow between mothers, CHVs, facilities, and the county; inefficient data management with paper records leading to data loss and repeats; barriers to proper maternal health services like long distances (50-100 km) and costs deterring check-ups; operational inefficiencies for workers with slow paper forms; and limited data for decision-making with no real time views.

The key question is, how do we build a basic system that joins all? One where mothers check signs on phone, CHVs update from spots/homes, and doctors advise online? And make sure regular CHV visits to cut child deaths and miscarriages?

1.3 Objectives

1.3.1 General Objective

To develop Boresha-Mama, a mobile and web-based maternal healthcare system for Trans-Nzoia County using technology to link mothers, CHVs, health facilities and county officials and make them more connected in providing maternal healthcare, lower child deaths and miscarriages via regular checks.

1.3.2 Specific Objectives

- I. To build a mobile application for mothers to track pregnancy progress, view due dates, receive health tips and reminders.
- II. To create tools specifically for CHVs using a mobile application which enable CHVs to enter data on home visits, record vitals, screen risks, and sync information to facilities.
- III. To develop web dashboards for health facilities to manage patient records, appointments, and

referrals.

IV. To implement county-level dashboards for viewing aggregated reports, maps, and planning metrics.

V. To pilot and evaluate the system in Kiminini constituency, Trans-Nzoia County, to gather user feedback and ensure usability.

1.4 Justification

Kenya is committed to improving maternal health outcomes nationwide. In Trans-Nzoia County, a significant number of women die from preventable causes—such as postpartum hemorrhage or untreated infections—due to limited or delayed access to quality care. These challenges also contribute to higher rates of neonatal deaths and miscarriages. The government has set ambitious targets to substantially reduce these figures by 2030, in line with the United Nations Sustainable Development Goals (particularly SDG 3 on good health and well-being). However, achieving these goals remains difficult without appropriate technological interventions and tools.

Boresha-Mama is designed to deliver meaningful savings in both time and financial resources. Expectant mothers will need to make fewer long journeys to health facilities, Community Health Volunteers (CHVs) will be able to complete their tasks more efficiently, and healthcare providers will be better positioned to offer timely support. By promoting consistent and scheduled CHV home visits, the system enables early detection and intervention for complications, helping to lower rates of child mortality and miscarriage. The platform is built to run on mobile phones, a device that is now widely accessible across Kenya. Affordable smartphones are increasingly available through flexible payment plans such as “lipa mdogo mdogo,” allowing users to acquire them gradually while paying in small, manageable instalments.

Furthermore, Boresha-Mama ensures that maternal health records are securely stored in the cloud, eliminating the risk of lost or damaged paper documents. It automatically sends appointment and visit reminders to both mothers and CHVs, improving adherence to care schedules. During public health emergencies or disease outbreaks, the system supports remote monitoring and consultation, reducing the need for in-person visits. This approach aligns closely with the Ministry of Health’s vision for digital health transformation in Kenya and provides a scalable model that can be expanded to other counties in the future.

By enabling earlier, more regular, and more consistent interaction with maternal health services, Boresha-Mama directly supports the reduction of preventable maternal and newborn deaths and illnesses. It strengthens the early recognition and management of high-risk pregnancies, eases the economic burden on families by minimising unnecessary travel for routine check-ups, and enhances overall efficiency—saving valuable time for both mothers and frontline health workers.

1.5 Scope of the Study

IN-SCOPE (What the system will do):

- i) My Boresha Mama maternal healthcare system will allow mothers in Trans-Nzoia County to register and create accounts to track and record their pregnancy progress, view timelines, receive reminders, and access health tips.
- ii) My Boresha Mama maternal healthcare system will allow mothers in Trans-Nzoia County to book appointments, check symptoms using the mobile app features, and locate nearest facilities on

maps.

iii) My Boresha Mama maternal healthcare system will allow mothers in Trans-Nzoia County to perform basic health watch (e.g., weight and blood pressure notes) by entering data during self-monitoring or CHV visits.

iv) My Boresha Mama maternal healthcare system will allow CHVs to record home visits offline, sync data, initiate referrals, and update vitals in real-time when connected.

v) My Boresha Mama maternal healthcare system will allow health facilities and county officials to view dashboards, generate reports, and plan maternal health services.

Out-of-Scope (What the system will not do):

i) My Boresha Mama maternal healthcare system will not allow non-Trans-Nzoia mothers to register and create accounts. It will not work for non-Trans-Nzoia residents to maintain focused county statistics and data integrity.

ii) My Boresha Mama maternal healthcare system will not allow joining with national health care systems e.g. Linda Mama/SHA at the initial phase. Integration can be considered in future expansions.

iii) My Boresha Mama maternal healthcare system will not include hardware development such as new medical devices for CHVs.

1.6 Limitations of the Proposed System

I. The system needs internet, but some places have poor network signals. This can slow down use in remote areas.

II. Not every mother has access to a smartphone, even with lipa mdogo mdogo. Some may need basic phones for SMS only.

III. Language barrier exists as the system will use English and Swahili. Some users speak local dialects, which might confuse them.

1.7 Project Risks and Mitigation

Risk: Low technological literacy among rural users.

Mitigation: Simple interfaces in English/Swahili and comprehensive training workshops.

Risk: Data security breaches.

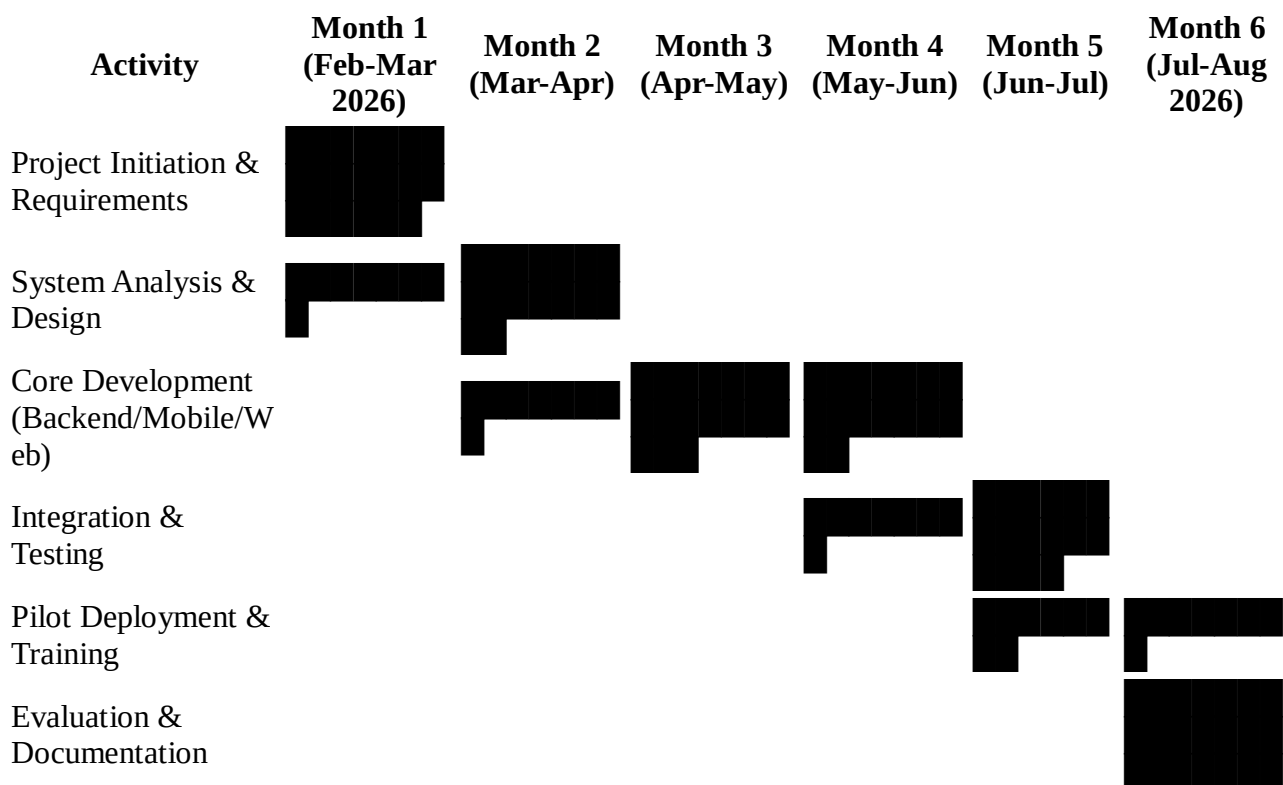
Mitigation: Encryption, role-based access, and compliance with Kenya Data Protection Act 2019.

Risk: Delays in county approvals.

Mitigation: Early stakeholder engagement starting in Month 1.

Risk: Cost overruns.

Mitigation: Use of free/open-source tools (e.g., React Native, PostgreSQL) and strict budgeting.



1.9 Budget and Resources

ITEM	COST (KSH)	DESCRIPTION	SPECIFICATION	UNIT
Laptop	40,000	For coding, testing, and documentation	Intel i5, 8GB RAM, 256GB SSD, Windows 11	1
Cloud Hosting/Server	25,000	Backend hosting and database	AWS/EC2 + PostgreSQL (6 months)	Subscription
Smartphones (Test)	30,000	Real-device testing for mobile apps	Android 12+, 4GB RAM, 64GB storage	2
Internet/Data	5,000	Bundles for development and field testing	10GB/month high-speed	6 months
Travel	20,000	Visits to Trans-Nzoia for stakeholder meetings	Fuel, public transport	Lump sum
Training Materials	10,000	Printing, posters, workshops	A4 paper, markers	Set
Expert Consultation	20,000	Health informatics advice	Maternal health specialist	10 hours
Awareness	5,000	Posters and community sensitisation	Printed materials	500 pcs
Antivirus & Tools	5,000	Security and code repository	License + GitHub Pro	Annual
TOTAL	160,000			

CHAPTER TWO: LITERATURE REVIEW

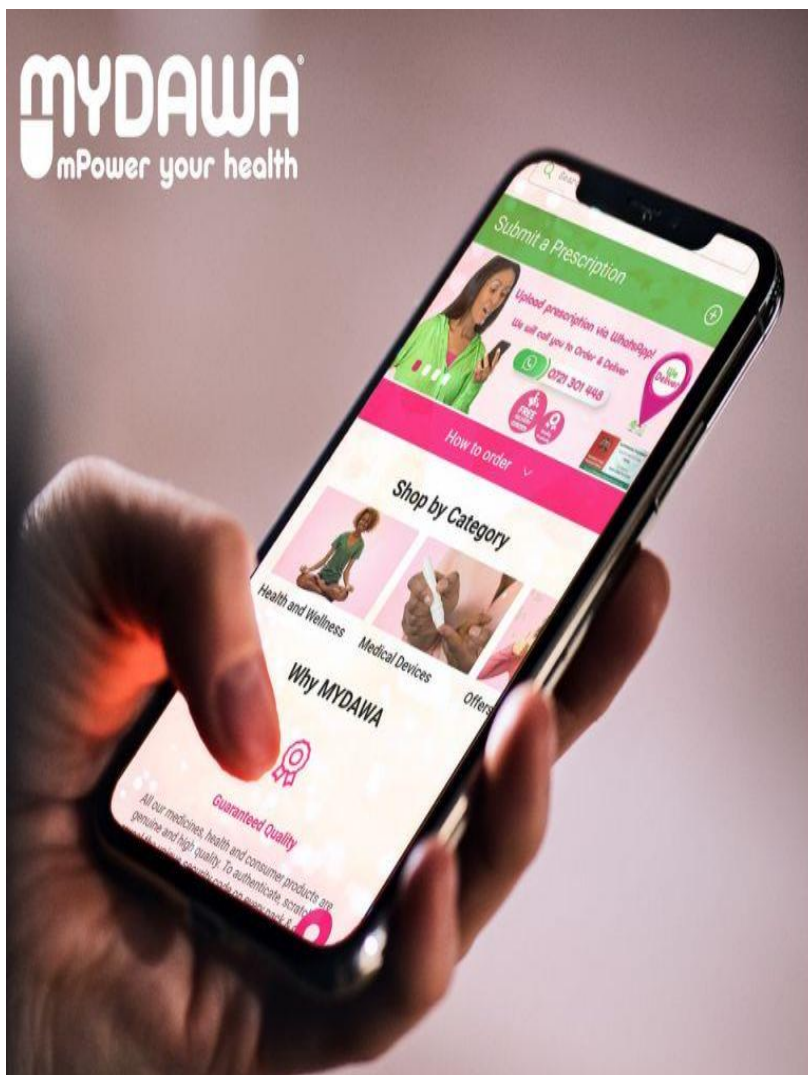
2.1 Introduction

This literature review intends to examine maternal healthcare systems in use in Kenya and around the world, including their strengths and drawbacks and compare them to my proposed Boresha Mama healthcare system.

2.2 Reviewed Similar Systems and Related Works

MyDawa (Kenya)

Developed in 2017 by Ion Kenya Ltd as Kenya's first online pharmacy and telehealth platform. It enables users to order medications, consult doctors via video, and receive home delivery. Good reviews for convenience and quality assurance; some users report occasional delivery delays. Overall user satisfaction is high (average 4.2/5 on app stores), particularly in urban areas.



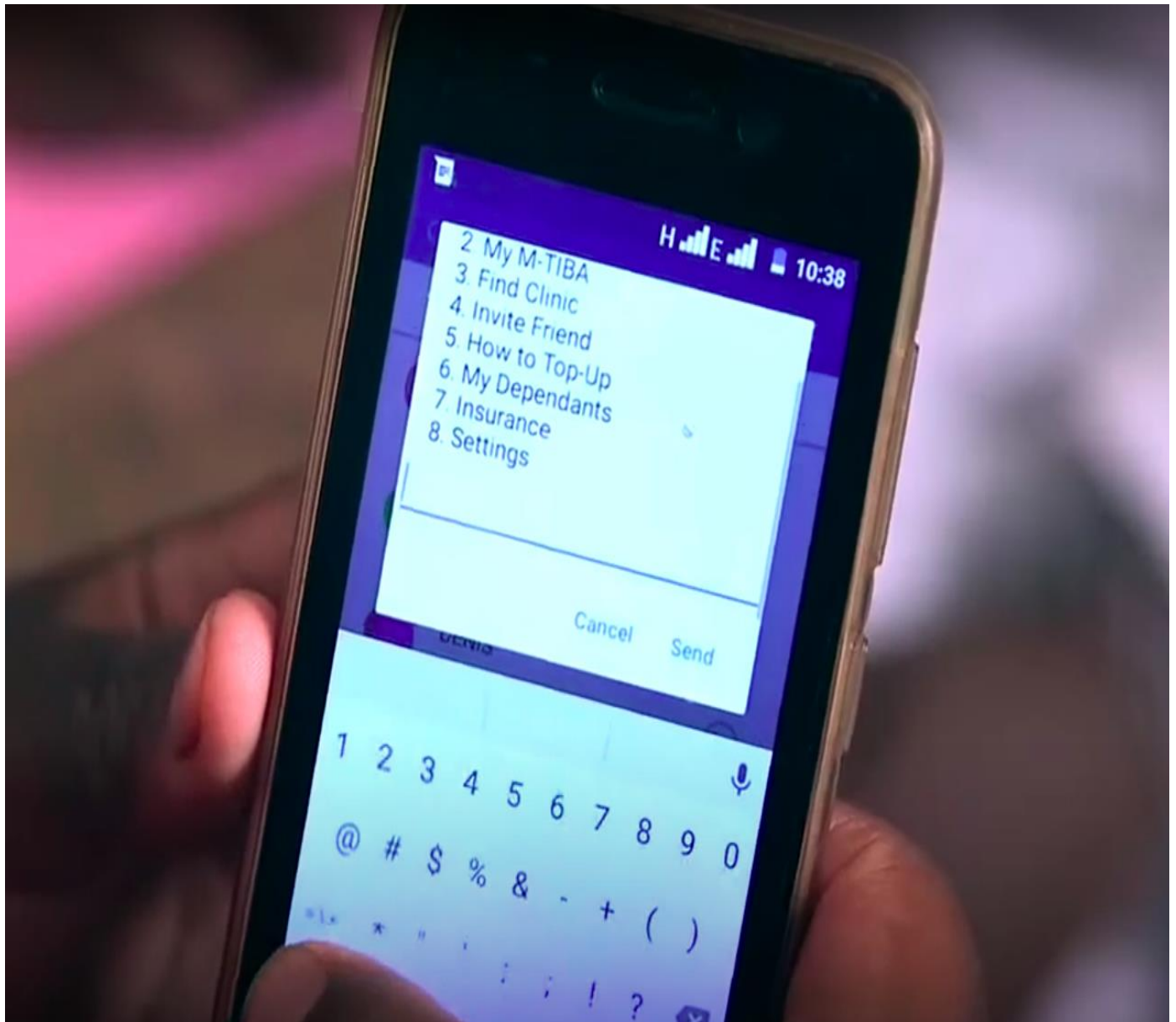
Available
on
iOS



M-TIBA (Kenya)

M-TIBA was launched in 2015/2016 by PharmAccess Foundation (now part of CarePay) as a mobile health wallet for saving and paying for healthcare in Kenya. It partners with health insurers and health providers for transparent claims.

Excellent reviews; Net Promoter Score of 60+, 96% reduction in claim processing time, high user satisfaction for accessibility on basic phones (M-TIBA, 2024).



Linda Mama (Kenya)

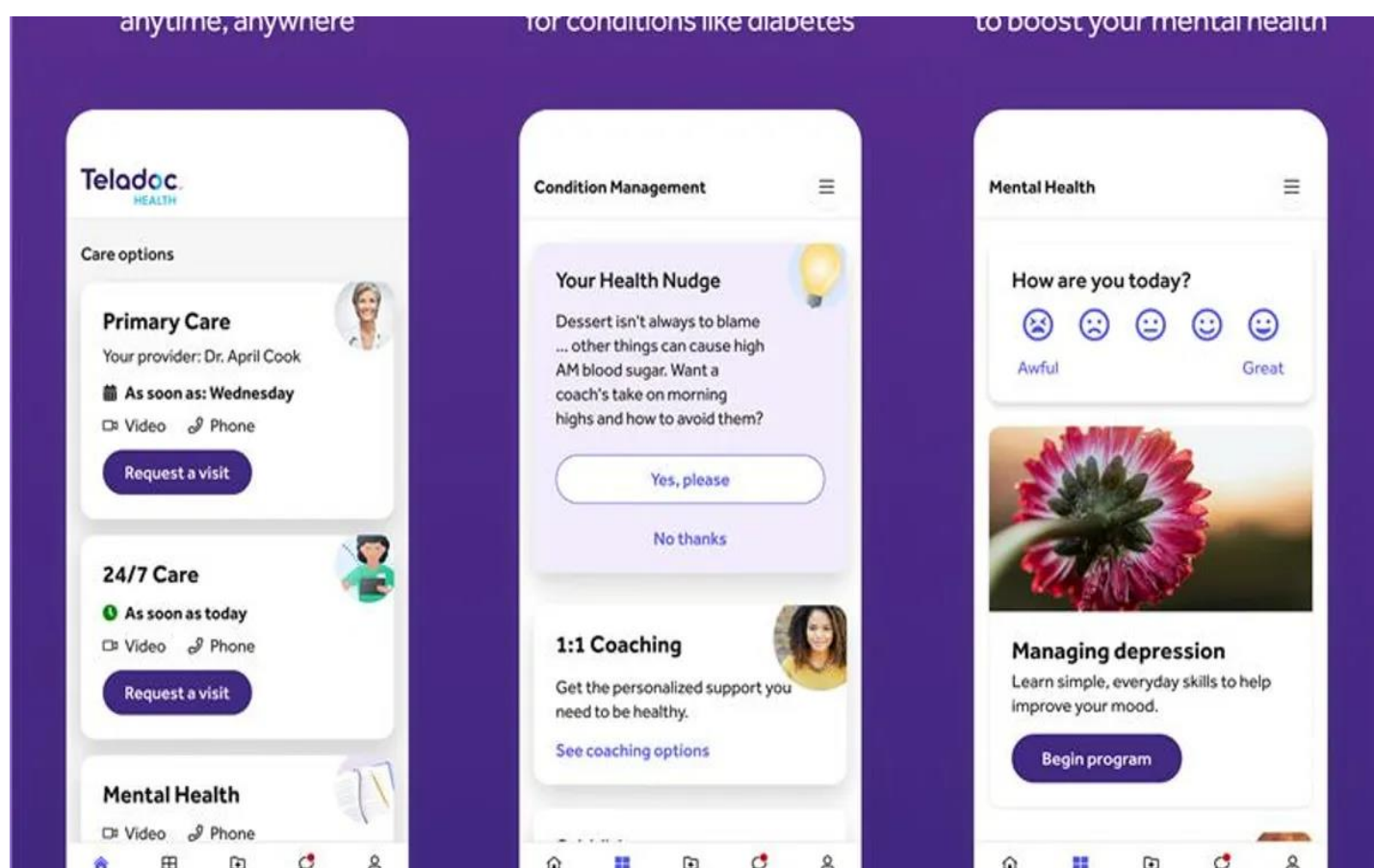
Launched 2017 under NHIF (integrated into SHA by 2024/2025). Provides free maternity services package.

Positive impact on utilisation (KDHS 2022); mixed reviews due to past payment delays to facilities, but high satisfaction among mothers for cost removal (Orangi et al., 2021).

Teladoc Global

Developed in 2002 in the USA as a telehealth platform for virtual consultations. Available in Kenya but limited by cost and connectivity.

Strong global reviews for accessibility; users satisfied with 24/7 care, though less adopted in rural Kenya due to internet requirements.



2.3 Tools and Methodologies Used in Reviewed Systems

- **MyDawa:** Mobile/web-based, centralised database, rule-based with some automation for order validation, semi-automated verification (prescription checks).
- **M-TIBA:** Mobile-based (USSD/app), centralised with real-time analytics, rule-based + AI for claims, automated validation and reporting.
- **Linda Mama:** Web-based portal for facilities (paper at community level), centralised NHIF/SHA database, rule-based, semi-automated claims.
- **Teladoc:** Web + mobile app, centralised, AI-assisted triage, automated scheduling and reporting.

2.4 Gaps in Existing Systems

- i. Most systems are for cities or general health- They focus on urban settings where internet is good.
- ii. They lack full connection for rural setups- No link between home visits and clinics.
- iii. There are no end-to-end from mother to county- Information doesn't flow all the way.
- iv. Paper records are still used leading to loss and delays.
- v. There is no/less focus on CHVs.
- vi. They don't have digital tools.
- vii. During COVID, remote care was needed but was not fully there. Systems weren't ready for no visits.

2.5 How This Project Fills the Gaps

Boresha-Mama joins all:

- i. Mothers mobile application for self-pregnancy track. They can see progress themselves.
- ii. CHVs update from the field- Quick data entry on site.
- iii. Clinics see real-time data- No waiting for papers.
- iv. County gets data for decisions and planning- See trends and plan better.
- v. It is made for Trans-Nzoia county, with offline mode. It fits local needs like poor network signals.

CHAPTER THREE: METHODOLOGY

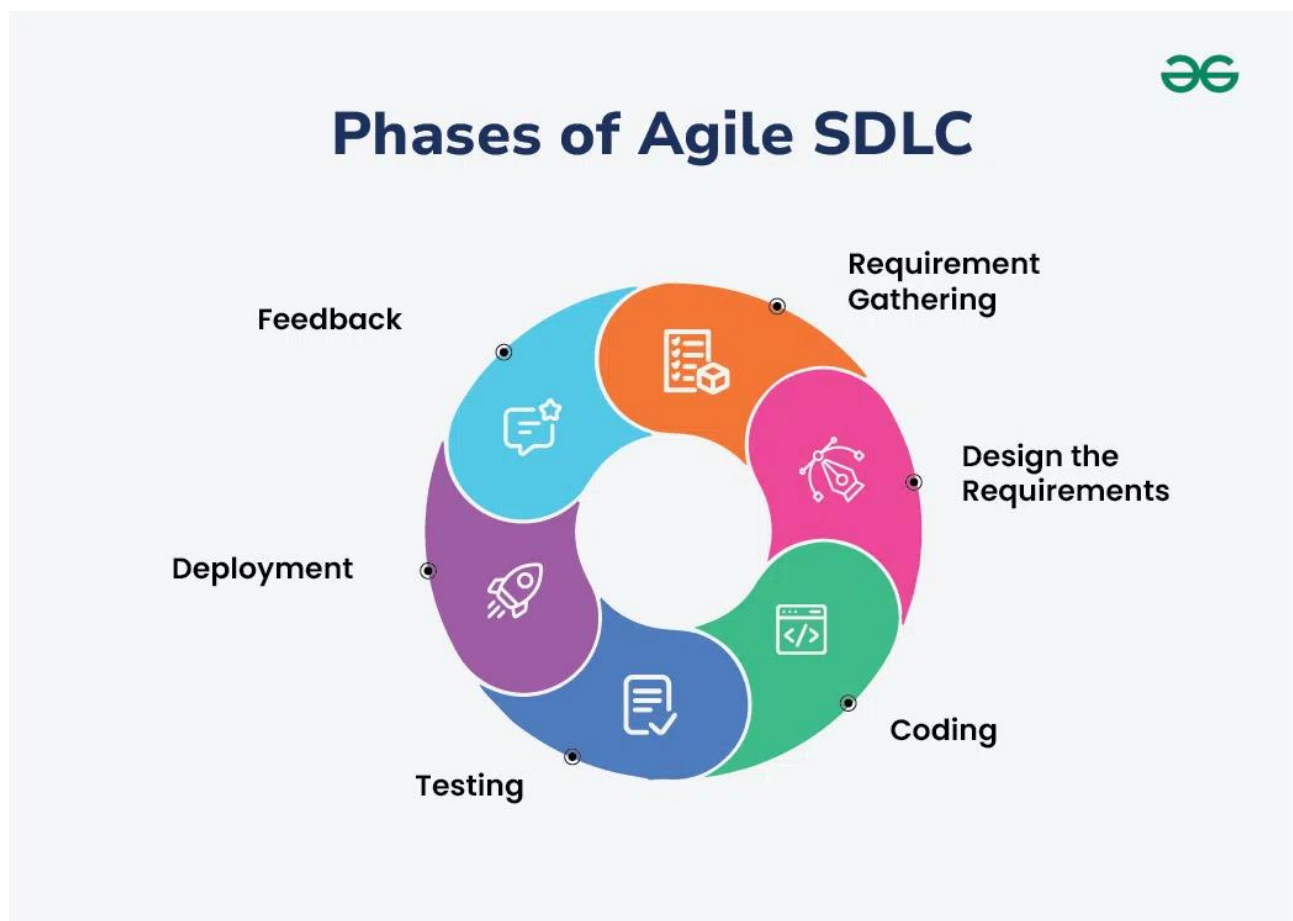
3.1 Introduction

This chapter will describe the techniques, methods, and strategies that will be used in the development of my Boresha-Mama system. In my study I will adopt the following research approach: mixed methods (qualitative interviews + quantitative questionnaires). In the chapter I will also outline the software development methodology and tools that will be used to implement my proposed system.

3.2 Selected Methodology: Agile Methodology

In this project, I choose the Agile Software Development Methodology. Agile can be defined as a Software Development Life Cycle (SDLC) methodology that emphasises iterative development, collaboration with stakeholders, and flexibility to respond to change.

I chose this methodology as it is suitable for the proposed system because it allows frequent user feedback from mothers and CHVs, enables incremental delivery of working features (e.g., offline sync first), and fits the small-team, time-constrained academic project.



3.3 Reasons for Choosing Agile

- i. Flexible for changes from user feedback- Can fix based on what people say.
- ii. Fast delivery of working parts- This helps get something usable quick.
- iii. Good for small team and limited time- I am the main developer and so it suits me.

3.4 Data Sources

Primary data sources: Interviews and questionnaires with stakeholders in Trans-Nzoia.

Secondary data sources: KNBS/MoH/WHO reports, peer-reviewed journals.

Formal definitions and examples:

- **Primary data:** Original data collected firsthand (e.g., responses from 50 mothers on travel distance).
- **Secondary data:** Existing published information (e.g., KDHS 2022 statistics on ANC coverage).

3.5 Data Collection Techniques

- **Interview:** A structured or semi-structured conversation to gather in-depth qualitative insights (direct verbal data collection from respondents). Questions will be discussed with supervisors. Sample interview question: “What challenges do you face accessing ANC services?”
- **Questionnaire:** A set of written questions for quantitative data. I will use structured (closed-ended) questionnaires for ease of analysis. Sample question: “On a scale of 1-5, how satisfied are you with current record-keeping at clinics?”

3.6 Resources / Materials Required

Hardware Specifications

- Laptop: Intel Core i5 12th Gen processor, 8GB RAM, 256GB SSD
- Windows 11
- Internet bandwidth 5Mbps+.

Software Specifications

- Android Studio (latest stable for React Native testing)
- Node.js v20 + Express
- PostgreSQL 16 (database)
- React Native 0.75 (mobile)
- React.js 18 (web dashboards)
- VS Code, GitHub repository, Avast Antivirus

CHAPTER FOUR: SYSTEM ANALYSIS AND REQUIREMENTS MODELING

4.1 Introduction

System analysis and design is the process of understanding the existing (current) system, identifying its deficiencies, gathering user requirements, and creating blueprints for a new or improved system that meets those needs effectively and efficiently.

This chapter presents a detailed analysis of the current maternal healthcare delivery processes in Trans-Nzoia County and outlines the requirements and models for the proposed Boresha-Mama system. The analysis employs structured techniques, including feasibility studies, data collection methods, fact-finding, and UML-based modeling tools, to ensure the proposed system is technically viable, economically justified, and operationally feasible.

4.2 Objectives of System Analysis and Design

The primary objectives of this chapter are:

- To critically examine the strengths and weaknesses of the current manual/paper-based maternal healthcare system.
- To elicit, document, and prioritise functional and non-functional requirements from stakeholders (mothers, CHVs, facility staff, and county officials).
- To model the proposed system using appropriate diagrammatic tools for clear visualisation of processes, data flows, entities, and interactions.
- To conduct a feasibility study to assess whether the Boresha-Mama system is practical and worthwhile.
- To provide a solid foundation for the subsequent development, implementation, and testing phases.

4.3 Feasibility Study

A feasibility study evaluates whether the proposed system is practical from multiple perspectives before committing resources.

4.3.1 Technical Feasibility

The proposed system is technically feasible because:

- It uses mature, open-source technologies (React Native for cross-platform mobile apps, React.js for web dashboards, Node.js + Express for backend, PostgreSQL for database).
- Offline functionality is supported via local storage (e.g., SQLite/Redux Persist in React Native) with background sync, addressing poor network issues in rural Trans-Nzoia.
- Hardware requirements are minimal (smartphones with Android 8+ or basic laptops for web access).
- Integration with SMS gateways (e.g., Africa's Talking) for reminders is straightforward and low-cost.
- No advanced AI or hardware is required initially; the system relies on rule-based logic for risk screening and reporting.

- Potential challenges (e.g., device compatibility) are mitigated through progressive web app (PWA) fallback and testing on low-end devices.

4.3.2 Economic Feasibility

The project is economically feasible with a total estimated cost of KSH 160,000 (detailed in Chapter 1.9).

Benefits include:

- Reduced travel costs for mothers (estimated savings of KSH 500–2,000 per ANC visit avoided through remote monitoring/reminders).
- Improved efficiency for CHVs and facilities (faster data entry/sync reduces administrative time by 40–60%).
- Potential reduction in preventable maternal/neonatal deaths, aligning with national health savings (per KDHS 2022, high ANC coverage correlates with lower mortality).
- Low ongoing costs (cloud hosting ~KSH 4,000/month after pilot).
- Cost-benefit analysis shows positive ROI within 12–18 months post-pilot, especially if scaled.

4.3.3 Operational Feasibility

The system is operationally feasible because:

- Stakeholders (mothers, CHVs) are already familiar with basic mobile phones via M-Pesa and similar apps.
- Interfaces will use simple English/Swahili, icons, and voice prompts where possible.
- Training workshops (planned in Month 6) will build capacity.
- The system supports existing workflows (e.g., CHV home visits) while digitising them.
- Acceptance is high due to direct benefits (e.g., reminders reduce missed visits). Resistance risks (e.g., from older users) are mitigated through participatory design and pilot feedback.

4.4 System Analysis Tools

The following UML and process-modeling tools are used to represent the system visually.

4.4.1 Entity-Relationship Diagram (ERD)

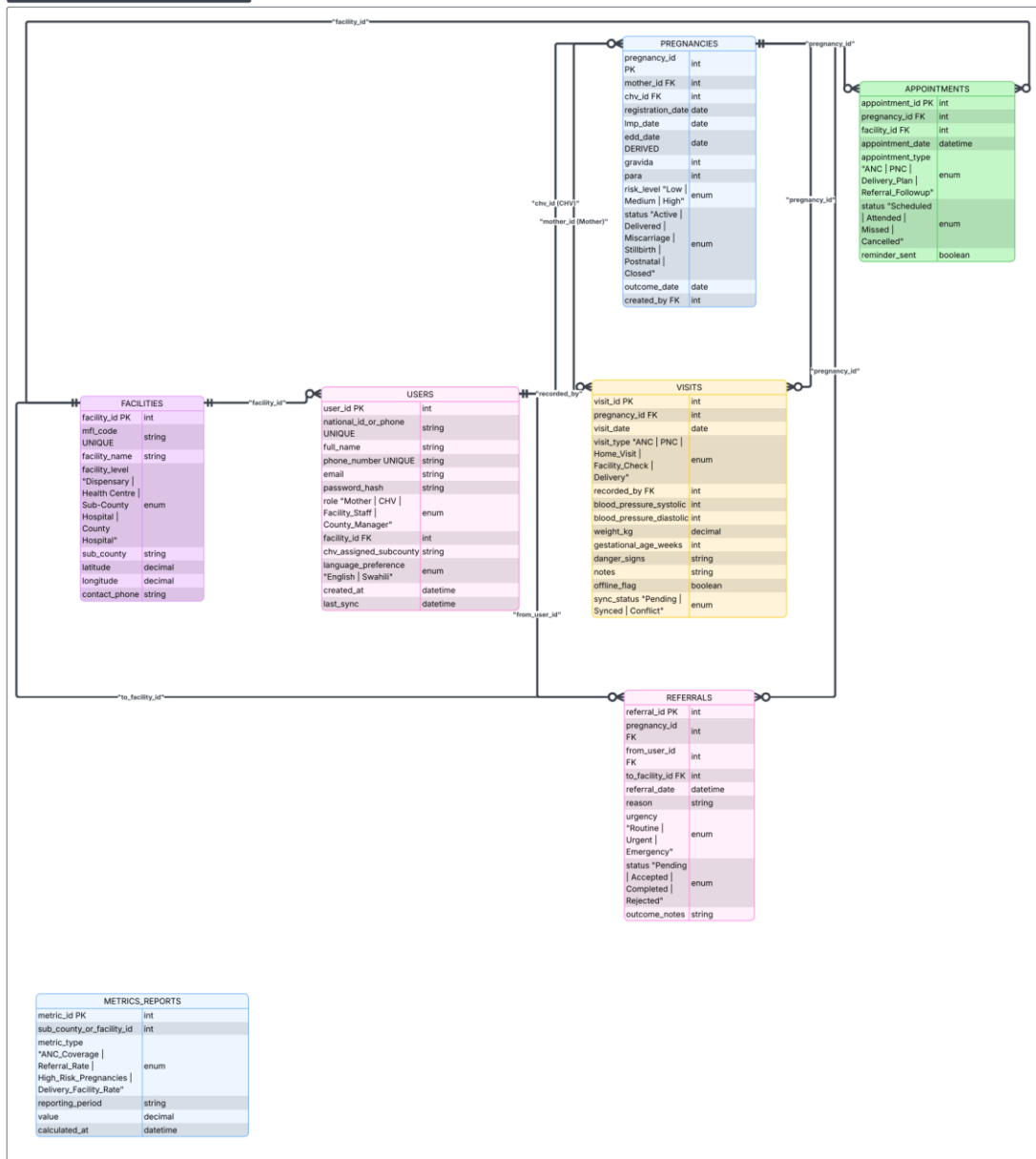
An **Entity-Relationship Diagram (ERD)** is a high-level conceptual data model that visually represents the entities (objects or concepts of interest in the domain), their attributes (properties), and the relationships among them. It serves as the foundation for logical and physical database design, helping to ensure data integrity, reduce redundancy, and support efficient querying. ERDs use standard notations such as Chen's or Crow's Foot (the latter is used here for clarity in cardinality).

The **Boresha-Mama ERD** is designed at the **conceptual level** to capture the core domain of maternal healthcare coordination in rural Trans-Nzoia County. It supports:

- Pregnancy lifecycle tracking (from registration to postnatal)

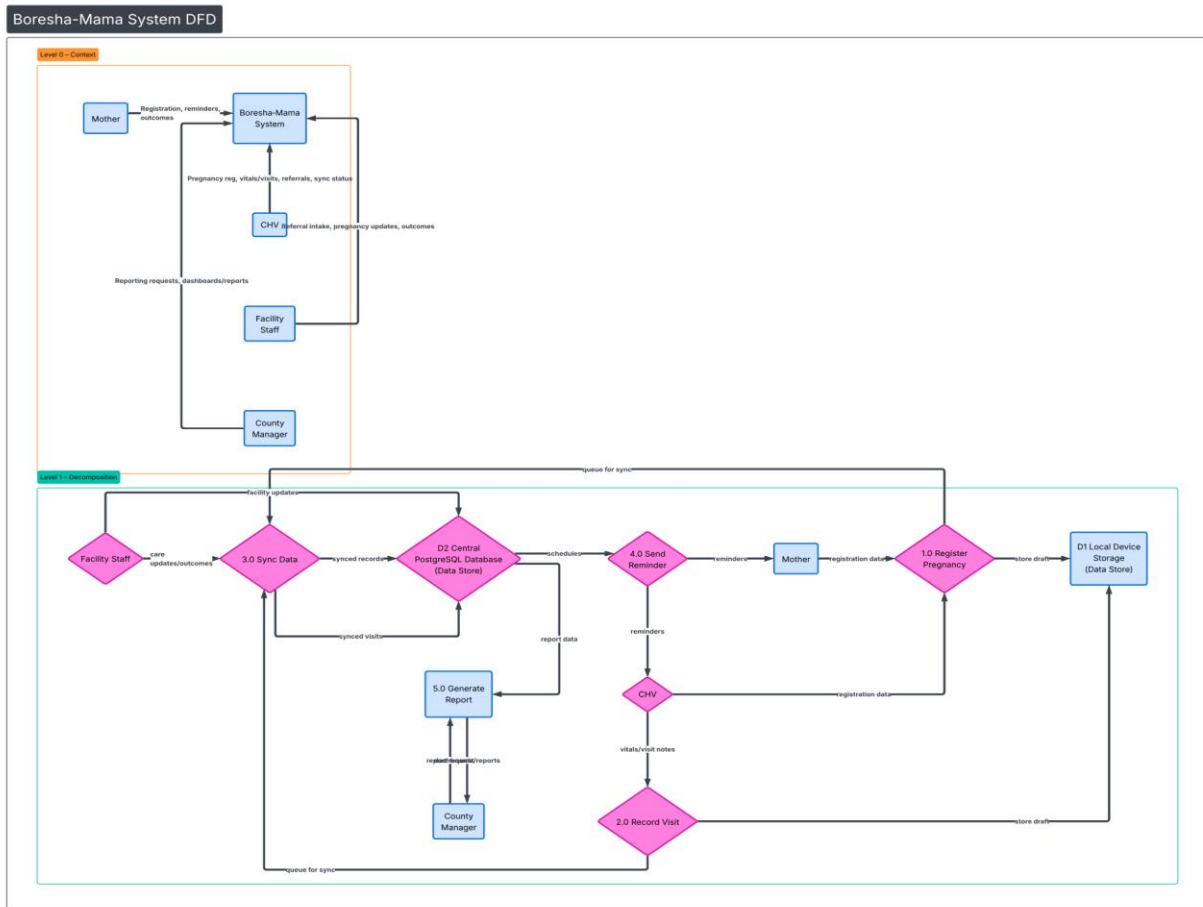
- Community-based monitoring by CHVs
- Facility-level care and referrals
- County-level aggregated reporting and planning
- Offline data capture with later synchronization
- Role-based data access (mothers see only their data; CHVs see assigned mothers; facilities see referrals; county sees anonymized aggregates)

Boresha-Mama ERD (Conceptual)



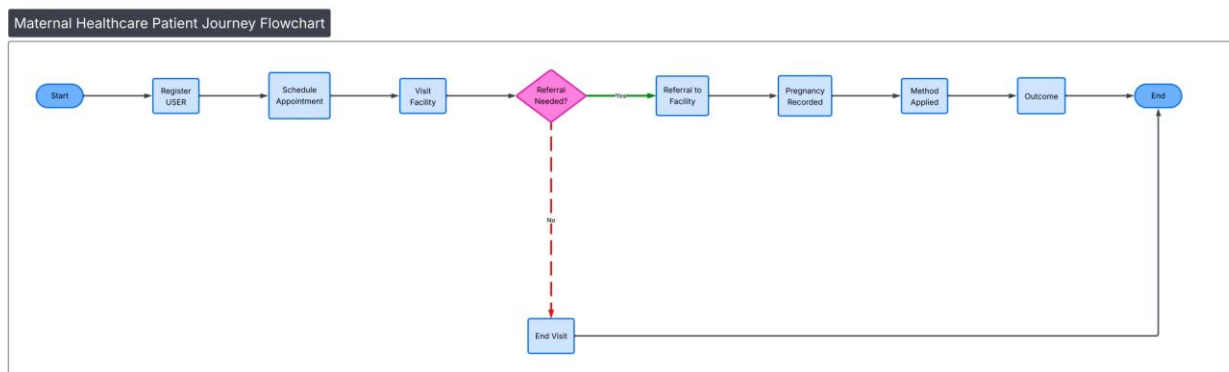
4.4.2 Data Flow Diagram (DFD)

A **Data Flow Diagram (DFD)** graphically represents how data moves through a system, showing processes, data stores, external entities, and data flows (no time sequence).



4.4.3 Flowchart

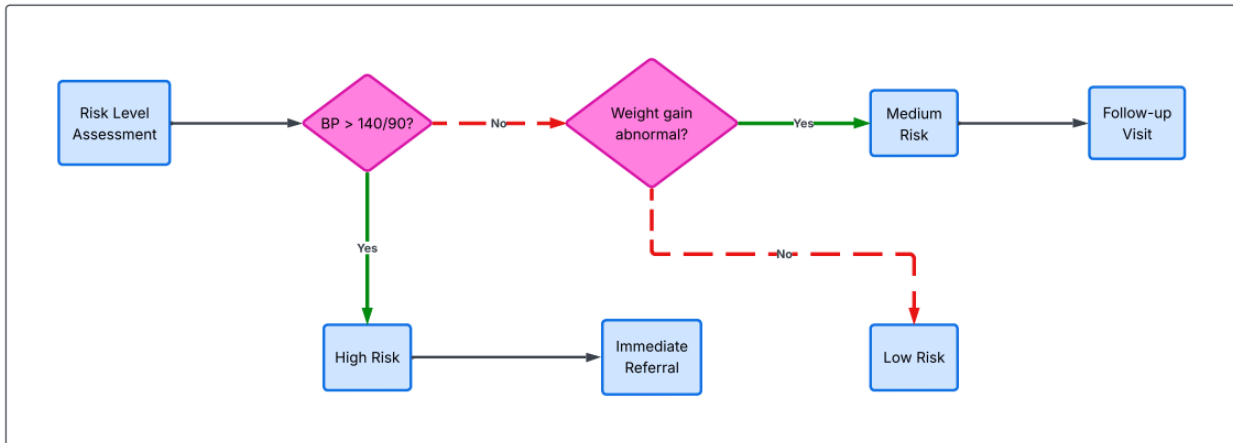
A **flowchart** is a diagrammatic representation of an algorithm or process using standard symbols (e.g., oval for start/end, rectangle for process, diamond for decision).



4.4.4 Decision Tree

A **decision tree** is a tree-like model showing decisions and their possible consequences, including chance event outcomes and resource costs.

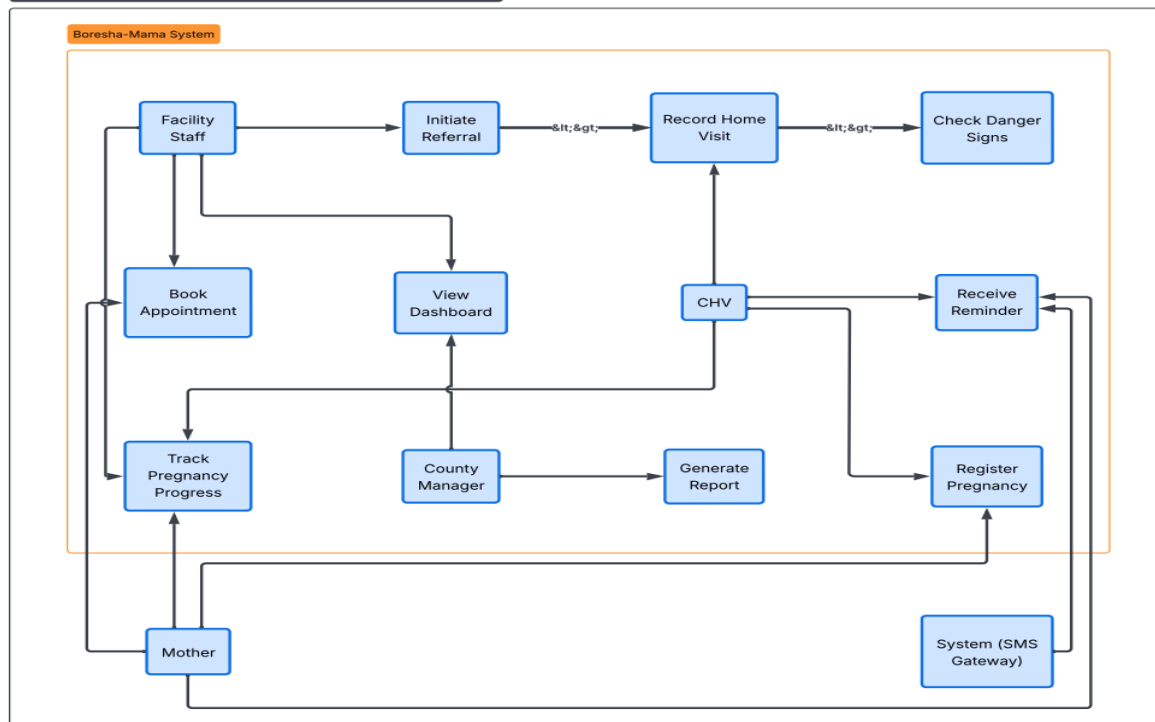
Pregnancy Risk Classification Decision Tree



4.4.5 Use Case Diagrams

A **use case diagram** (part of UML) illustrates the interactions between actors (users) and the system via use cases (functionalities).

Boresha-Mama Use Case Diagram



4.5 System Investigation

4.5.1 Introduction

System investigation involves gathering detailed facts about the current system to identify problems and user needs.

4.5.2 Data Collection

Methods used:

- **Interviews:** Structured/semi-structured discussions with ~20 stakeholders (mothers, CHVs, nurses, county officers) to explore experiences qualitatively.
- **Questionnaires:** Structured (closed-ended Likert-scale + some open) distributed to ~100 respondents for quantitative data on challenges (e.g., travel distance, record loss frequency).
- **Observation:** Direct watching of CHV home visits and clinic processes.
- **Document Review:** Analysis of Mwananchi cards, ANC registers, KNBS/MoH reports.

4.5.3 Fact Recording

Findings recorded via: audio recordings (with consent), field notes, questionnaire summaries, and thematic coding. Key facts: 70% of mothers travel >10 km for ANC; paper records lost in 40% of cases; CHVs spend >30% time on manual documentation.

4.6 System Analysis

4.6.1 Functional Requirements

- The registration module should ensure secure user registration of mothers, CHVs, facility staff, and county officials through verified account sign-up (e.g., phone OTP + CHV/facility confirmation).
- The mother mobile app should ensure pregnancy tracking, display timelines (LMP to EDD), provide personalised health tips, and send push/SMS reminders.
- The CHV mobile app should ensure offline recording of home visits, entry of vitals (BP, weight), danger sign screening, referral initiation, and automatic sync when online.
- The facility web dashboard should ensure real-time viewing of assigned pregnancies, management of appointments/bookings, update of referral outcomes, and basic reporting.
- The county web dashboard should ensure aggregated KPI visualisation (e.g., ANC coverage maps, referral rates), report generation/export (PDF/CSV), and user/facility management.

4.6.2 Non-Functional Requirements

- The system should ensure high performance with response times <3 seconds for online operations and full offline CRUD support.
- The user interface should ensure usability with intuitive design, large buttons, Swahili/English toggle, and accessibility for low-literacy users (e.g., icons + voice hints).
- The system should ensure security through JWT authentication, role-based access control (RBAC), data encryption (at rest/transit), and audit logs, complying with Kenya Data Protection Act 2019.
- The system should ensure reliability with 99% uptime (cloud-hosted), automatic backups, and conflict resolution during sync.
- The system should ensure scalability to support 10,000+ users in Trans-Nzoia and portability across Android/iOS via React Native.

CONCLUSION

Boresha-Mama will solve real problems in maternal health in Trans-Nzoia. It connects people with simple technology to save lives. It reduces travel, keeps records safe, and helps early detection of maternal health problems. The plan is clear, low-cost, and matches Kenya's health goals. After pilot success, it can help more counties countrywide. This project is significant for its potential to create tangible impact across multiple domains: clinical by reducing deaths, socio-economic by saving money, operational by streamlining work, and academic by sharing knowledge. The solution is designed for scalability and adaptation.

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APPENDICES

Appendix A: Questionnaire and Interview Guide

Sample Questions for Mothers:

1. How often do you attend antenatal care visits?
2. What challenges do you face in accessing maternal healthcare services?
3. How do you currently track your pregnancy progress?

Sample Questions for CHVs:

1. How do you record information during home visits?
2. What challenges do you face in sharing data with health facilities?
3. How would a mobile app improve your work?

Sample Questions for Health Workers/Officials:

1. How do you currently manage maternal health records?
2. What challenges exist in planning maternal health services?
3. How would real-time dashboards improve decision-making?

Appendix B: Sample Responses (to be completed post-pilot)

Mother's Response:

- “I attend antenatal visits only when I can afford transport.”
- “The biggest challenge is distance and poor roads.”
- “I track my pregnancy using a clinic card, but sometimes it gets lost.”

CHV's Response:

- “I record visits in notebooks, which take long to reach clinics.”
- “Sharing data is slow because records are paper-based.”
- “A mobile app would help me update information instantly.”

Health Worker's Response:

- “We manage records manually, which is time-consuming.”
- “Planning is difficult without up-to-date data.”
- “Dashboards would help us allocate resources better.”

Appendix C: PlantUML Code for Sequence Diagrams

@startuml

actor Mother

actor CHV

actor HealthFacility

actor CountyOfficial

Mother --> (Register Pregnancy)

Mother --> (Track Pregnancy Progress)

CHV --> (Record Home Visit)

CHV --> (Update Vitals)

HealthFacility --> (Manage Records)

CountyOfficial --> (View Reports)

(Register Pregnancy) --> (Database)

(Track Pregnancy Progress) --> (Database)

(Record Home Visit) --> (Database)

(Update Vitals) --> (Database)

(Manage Records) --> (Database)

(View Reports) --> (Database)

@enduml